India EV Aftersales Strategy

₹38,000 Cr
TAM by FY30

Identifying ₹680 Crore in Untapped Value
Across OEM Service Networks

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Delhi Technological University | 2027

Python | Power BI | Monte Carlo | Pandas

THE PROBLEM

India's EV service infrastructure is not built for what is coming

₹38K Cr

Market being missed

EV aftersales TAM reaches ₹38,000 Cr by FY30 — OEMs are capturing less than 20% of potential today

65%

Dealers losing money

65% of Tier-3 service centers operate below breakeven every month due to low throughput vs high fixed costs

32%

Warranty waste

32% of all warranty workshop visits are OTA-avoidable — costing ₹70 Cr annually in unnecessary repairs

This study identifies 3 value levers that unlock ₹680 Cr in aftersales EBITDA by FY27

APPROACH

Hypothesis-driven framework across 3 MECE value levers

Can India EV OEMs improve aftersales EBITDA by 8–12pp?

Parts Pricing

- H1a: EV parts underpriced vs ICE equivalents
- H1b: Battery module pricing leaves margin on the table
- H1c: Aftermarket cannibalization reduces OEM parts revenue

Service Network ROI

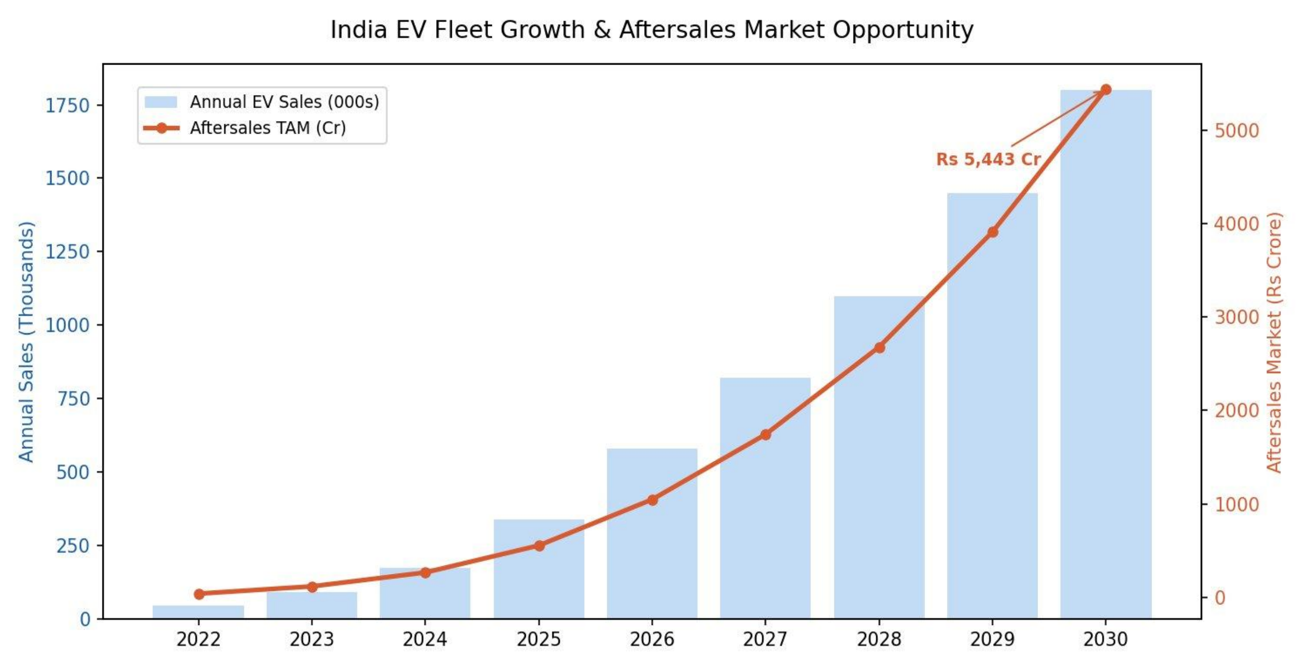
- H2a: Tier-3 dealer throughput below breakeven threshold
- H2b: Mobile vans have 2x ROI vs fixed centers
- H2c: OTA updates reduce 30–40% of workshop visits

Warranty Cost

- H3a: Reserves overestimate battery failure rates
- H3b: Telematics can reduce claims by 20–25%
- H3c: Climate-adjusted models need India recalibration

FINDING 1

India EV aftersales TAM reaches ₹38,000 Cr by FY30



₹38K Cr

Aftersales TAM FY30

64L+

Cumulative EV Fleet FY30

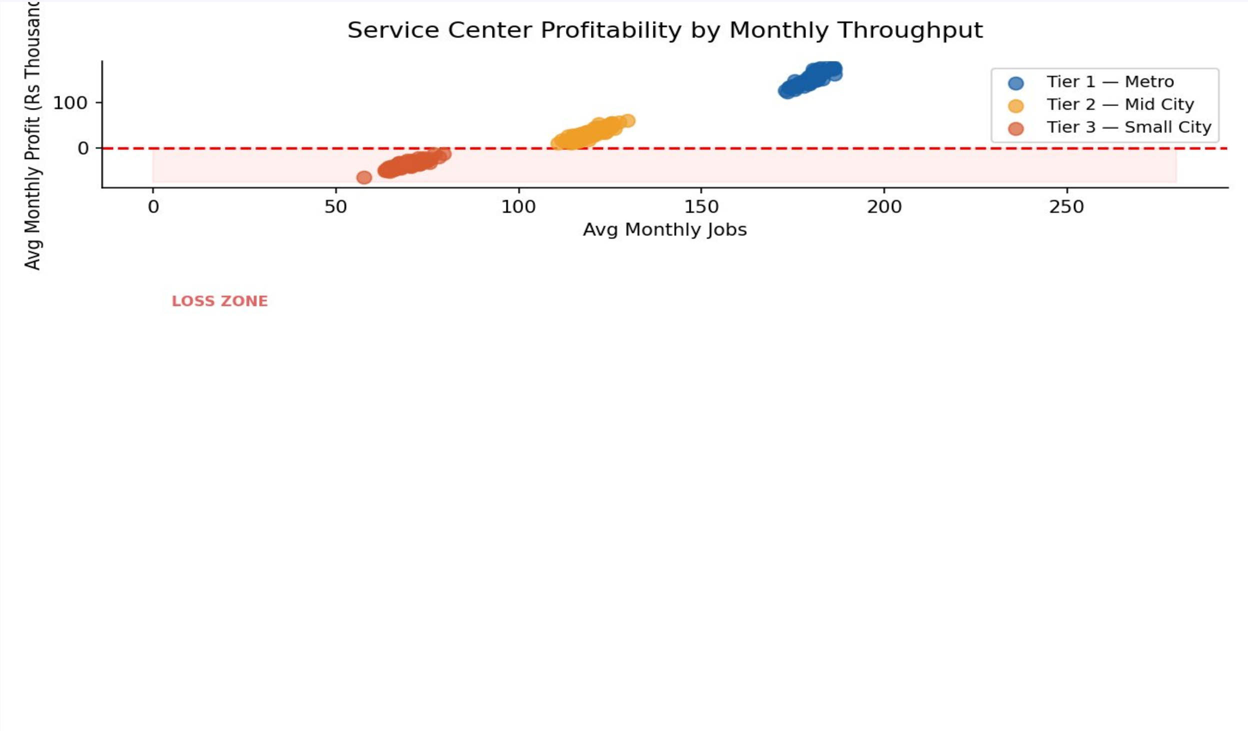
₹8,500

Avg spend / vehicle / year

Tata Motors (53% fleet share) has the single largest stake in this opportunity — every 1% improvement in aftersales margin = ₹380 Cr annual EBITDA impact

FINDING 2

65% of Tier-3 service centers are loss-making every month



Tier 1 — Metro

Actual jobs: 181 | Breakeven: 126

+₹3.2L/mo

Tier 2 — Mid City

Actual jobs: 120 | Breakeven: 116

+₹48K/mo

Tier 3 — Small

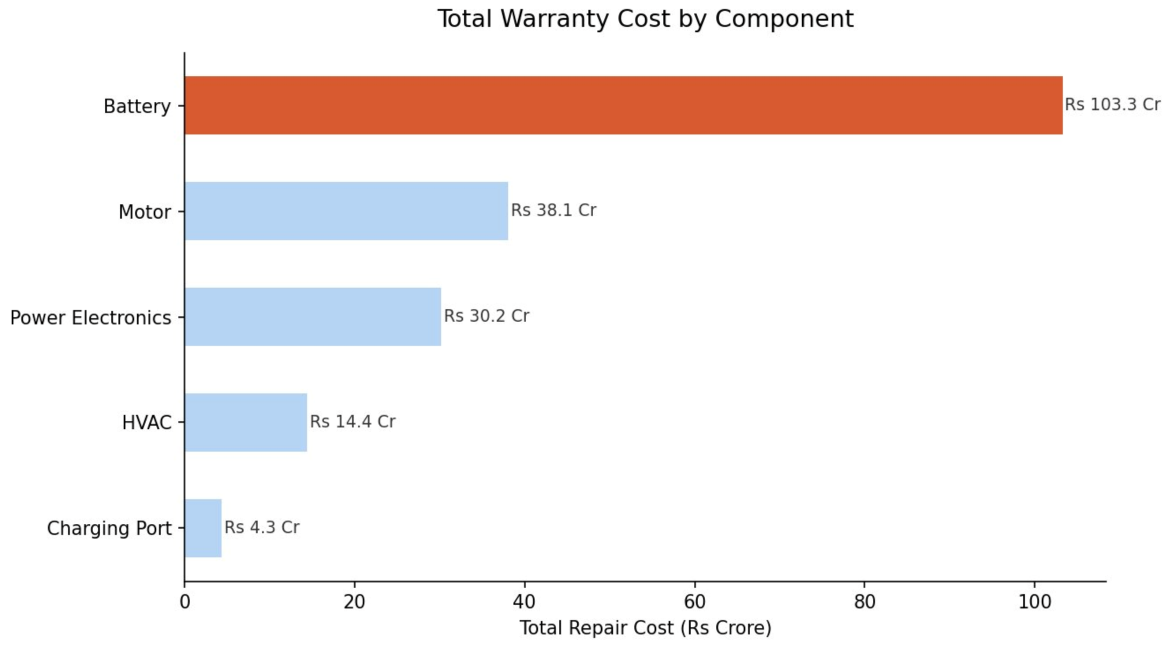
Actual jobs: 70 | Breakeven: 145

-₹82K/mo

Tier 3 needs 2x current throughput to survive

FINDING 3

Battery accounts for 43% of total warranty spend at ₹55K avg repair cost



₹103 Cr
Battery total warranty cost

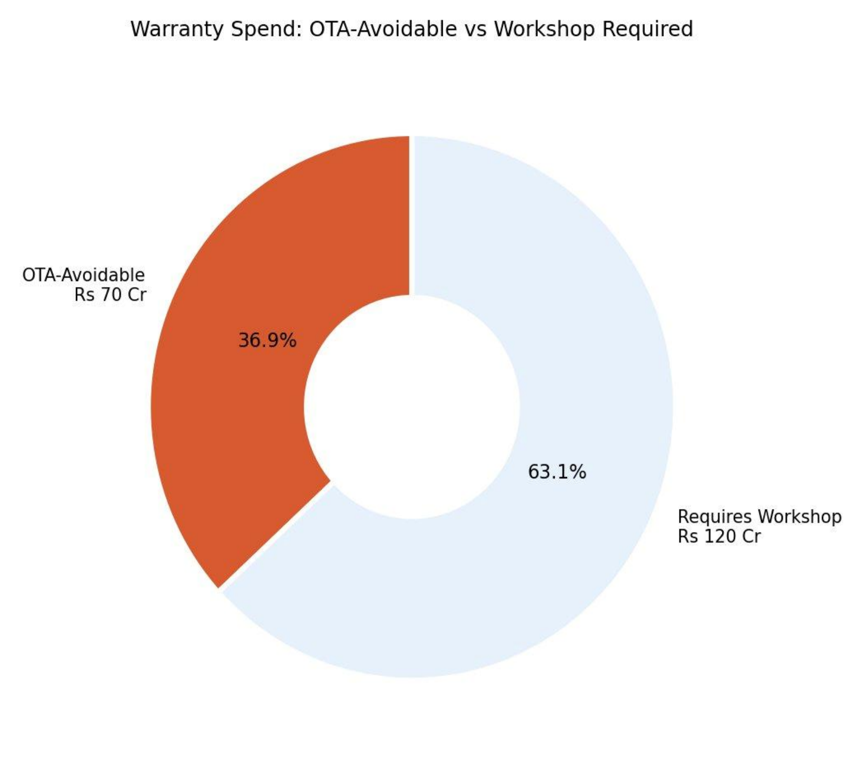
₹55K
Avg battery repair vs ₹8K charging port

43%
Battery share of total warranty spend

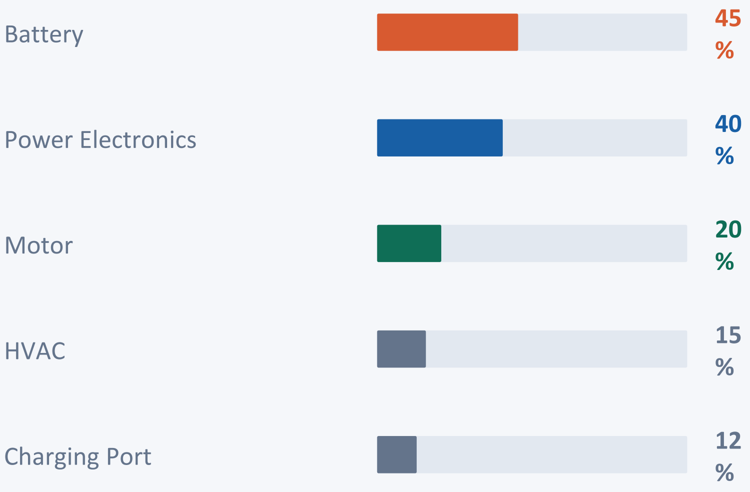
India-specific insight: Hot-Humid climate zones show higher battery failure rates than Temperate zones — current OEM warranty reserves use global benchmarks and overestimate India-specific risk by ~18–25%

FINDING 4

37% of warranty workshop visits are OTA-avoidable — saving ₹70 Cr annually



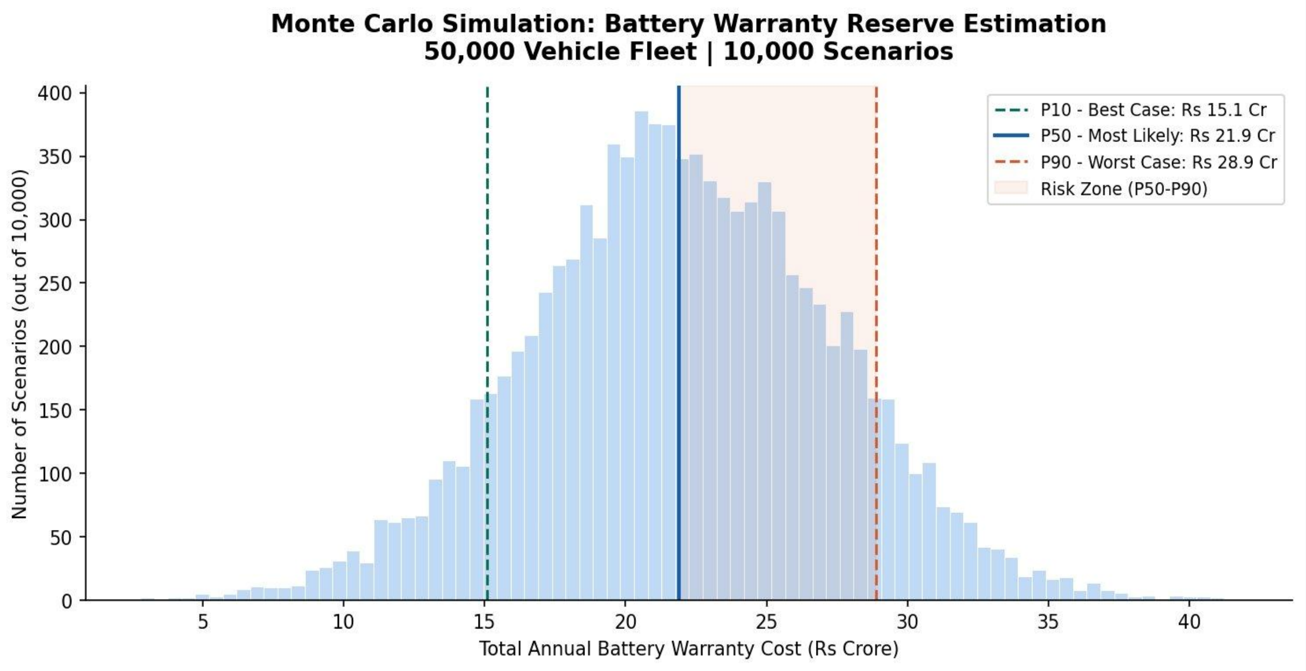
OTA avoidable rate by component



Tesla benchmark: 35–40% of service visits resolved via OTA. India OEMs currently: ~5%

MONTE CARLO

OEMs over-provision warranty reserves by 18–25% — simulation shows the real range



₹15.1 Cr

P10 — Best Case

₹21.9 Cr

P50 — Most Likely

₹28.9 Cr

P90 — Worst Case

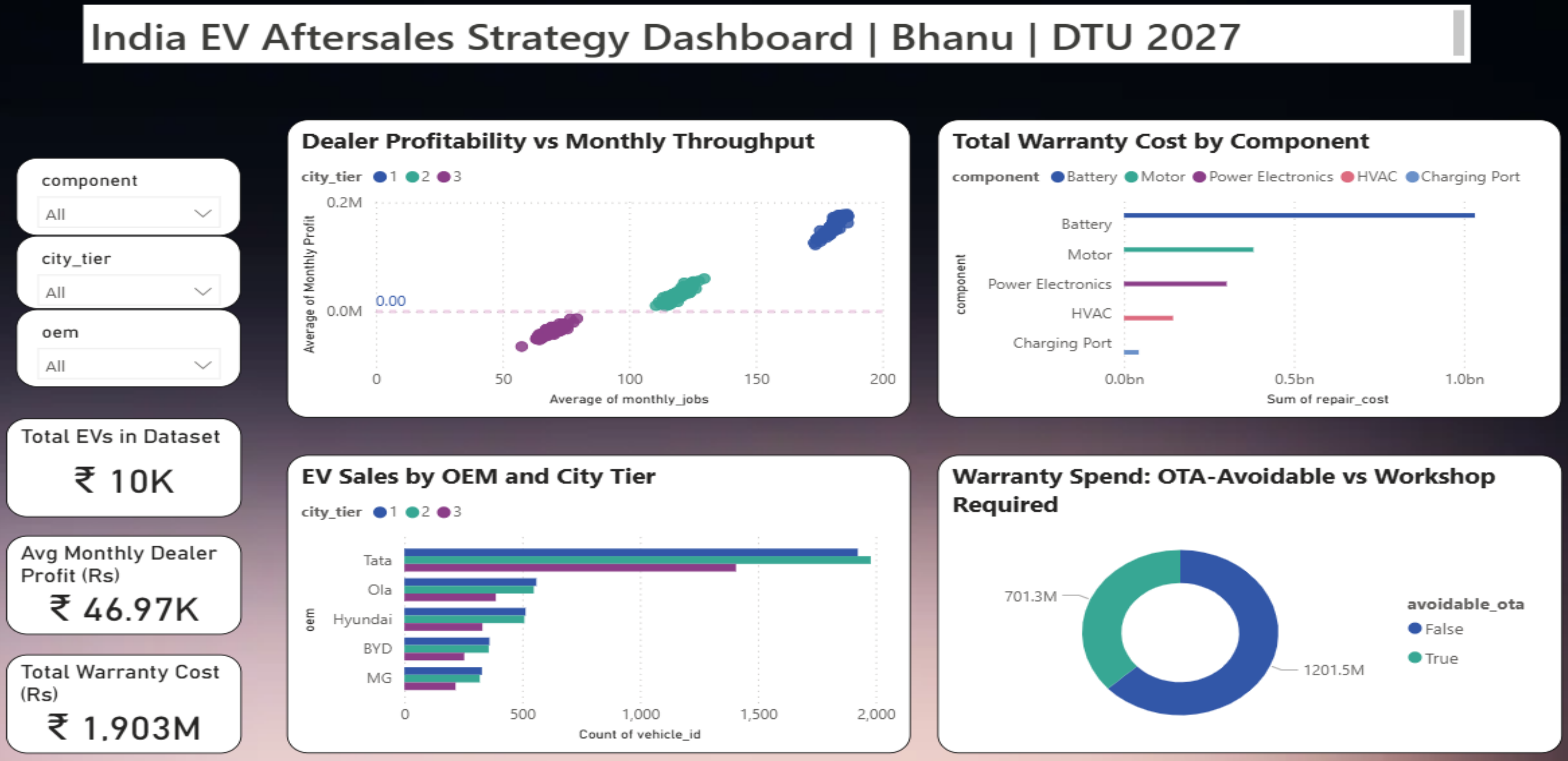
₹25 Cr

Recommended Reserve (P75)

Methodology: 10,000 simulations | failure rate N(8%, 2%) | repair cost lognormal(₹55K, ₹15K) | 50,000 vehicle fleet

POWER BI

Dashboard for India EV Aftersales Strategy



RECOMMENDATIONS

3 moves that unlock ₹680 Crore in aftersales EBITDA by FY27

01

Parts Pricing Optimization

Increase pricing on 40 high-velocity SKUs by 8–12% — still within consumer tolerance given EV complexity premium. No capital investment required.

₹480 Cr/yr

annual impact

DO NOW

0–6 months

02

OTA Software Deflection Program

Deploy proactive telematics-triggered OTA updates for Battery and Power Electronics. Battery (45% avoidable) and Power Electronics (40%) are the priority.

₹200 Cr/yr

annual impact

PLAN

6–18 months

03

Mobile Service Vans — Tier 3

Replace fixed Tier-3 service centers with mobile vans. Reduces fixed cost from ₹1.8L/month to ₹60K/month, closing the throughput gap without scaling footprint.

₹160 Cr/yr

annual impact

PLAN

12–24 months

COMBINED IMPACT: ₹680 Crore annual EBITDA improvement — a 2.1× increase from current estimated ₹320 Crore

APPENDIX

KPIs, Benchmarks & Methodology

KPI Metric	Definition	Model Output	Benchmark
Aftersales EBITDA Margin	Aftersales EBIT / Revenue	18–22% (India EV est.)	Toyota: 45–55%
Parts Gross Margin	(Revenue − COGS) / Revenue	28–32% currently	ICE benchmark: 35–40%
Dealer Breakeven Jobs	Min jobs to cover fixed costs	T1: 126 · T2: 116 · T3: 145	Industry: 120–180/month
Warranty Cost % Revenue	Warranty spend / Vehicle rev	India EV: ~4–5%	Global EV: 2.5–3.5%
OTA Deflection Rate	% visits resolved via software	32–45% (your data)	Tesla benchmark: 35–40%
Revenue per Service Outlet	Aftersales rev / service points	₹1.2–2.5 Cr/year	Maruti: ₹3–4 Cr/year

Tools & Stack

Python (pandas, numpy, matplotlib, scipy)

Power BI · Monte Carlo simulation · PowerPoint

Data Sources & Scale

10K EV sales · 7.2K service records · 50K warranty claims

Simulated from VAHAN fleet benchmarks + SMEV 2023–24